SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID – SIH25042**
- **Problem Statement Title- Identifying Taxonomy and Assessing Biodiversity form eDNA Datasets**
- **Theme- Miscellaneous**
- **PS Category- Software**
- **Team ID-**
- **Team Name (Registered on portal)- Winners and sons.**

Winners and sons

IDEA TITLE - Oceanus AI

- We made an AI based platform based of marine data across the world to help researchers identify DNA sequence faster and with high accuracy

- Traditionally it took 2-4 weeks from collecting water samples to determining the species with the DNA extracted.

- With our product we have reduced that time to < 1 week with major
- Time consuming part being DNA sequencing which has to be done in Labs

- No Login required user can directly visit the dashboard and can use the AI tool for research purpose

Tech Stack

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TECHNICAL APPROACH

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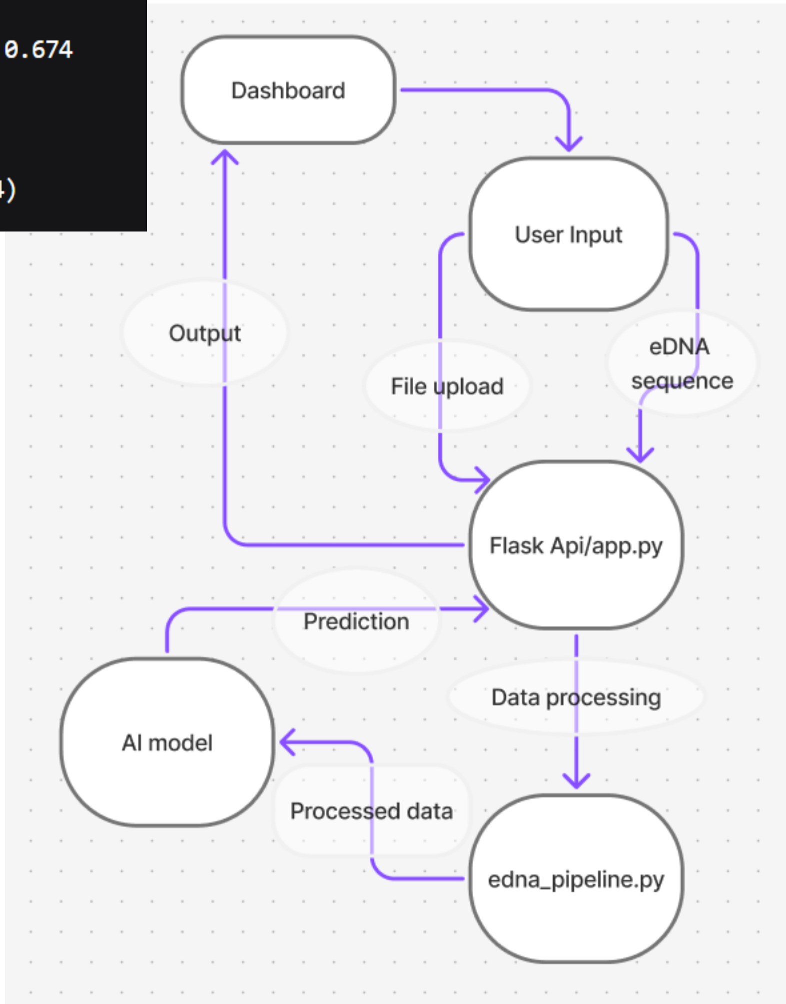
```
Best K-Means: 19 clusters, silhouette score: 0.594
Training DBSCAN...
Best DBSCAN: 6 clusters, silhouette score: 0.454
Training Hierarchical clustering...
Best Hierarchical: 2 clusters, silhouette score: 0.674

Analyzing results...

Best performing model: hierarchical (score: 0.674)
```

Frontend: React, Next.js, Typescript, Tailwind CSS
Backend: Python, Flask, Scikit learn
Deployment: Vercel, Vercel Blob

Using the dashboard is very easy user can simply enter the DNA sequence to identify if the file size is less or not many DNA sequence is available or can upload the file in various format like .csv, .fa, .fasta, .fas , etc to upload the DNA sequence

Which the API sends to the eda_pipeline.py for further processing or data and then evaluated by the unsupervised learning AI model of silhouette score of 0.674

The evaluated result is then displayed to the user on the dashboard



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FEASIBILITY AND VIABILITY

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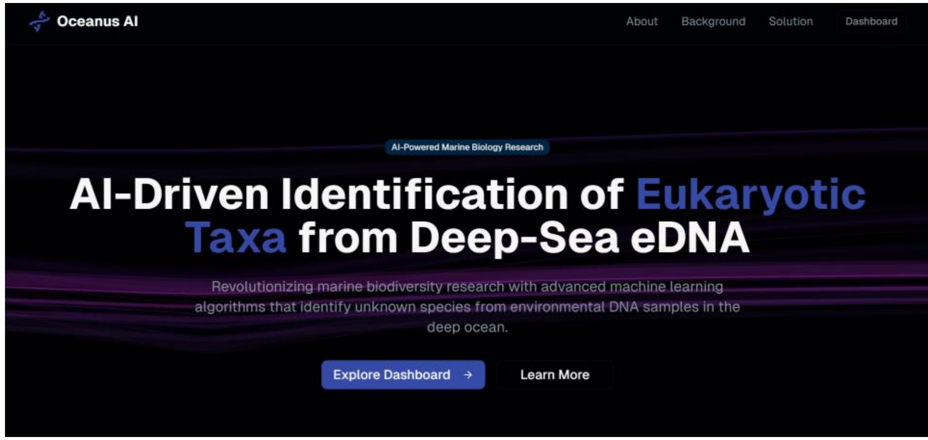
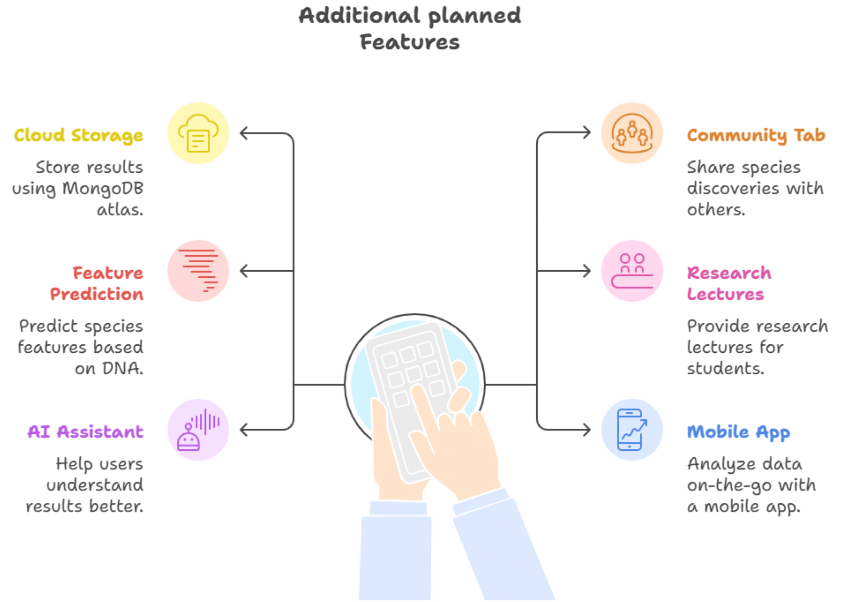
Oceanus AI is very easy and straight forward to use and support various input type for easy access backed by a very capable AI model for clustering and differentiating species in classified and novel species

But it lack many features which we would like to implement to make the project even better and user/community friendly like:

- Cloud based result storage using MongoDB atlas
- Community tab to share your work of what species and where user discovered that
- Feature prediction for species based of their DNA, RNA structure
- Research lectures for students
- AI assistant to help understand the results better
- Mobile app for on-the-go analysis

And many more





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IMPACT AND BENEFITS

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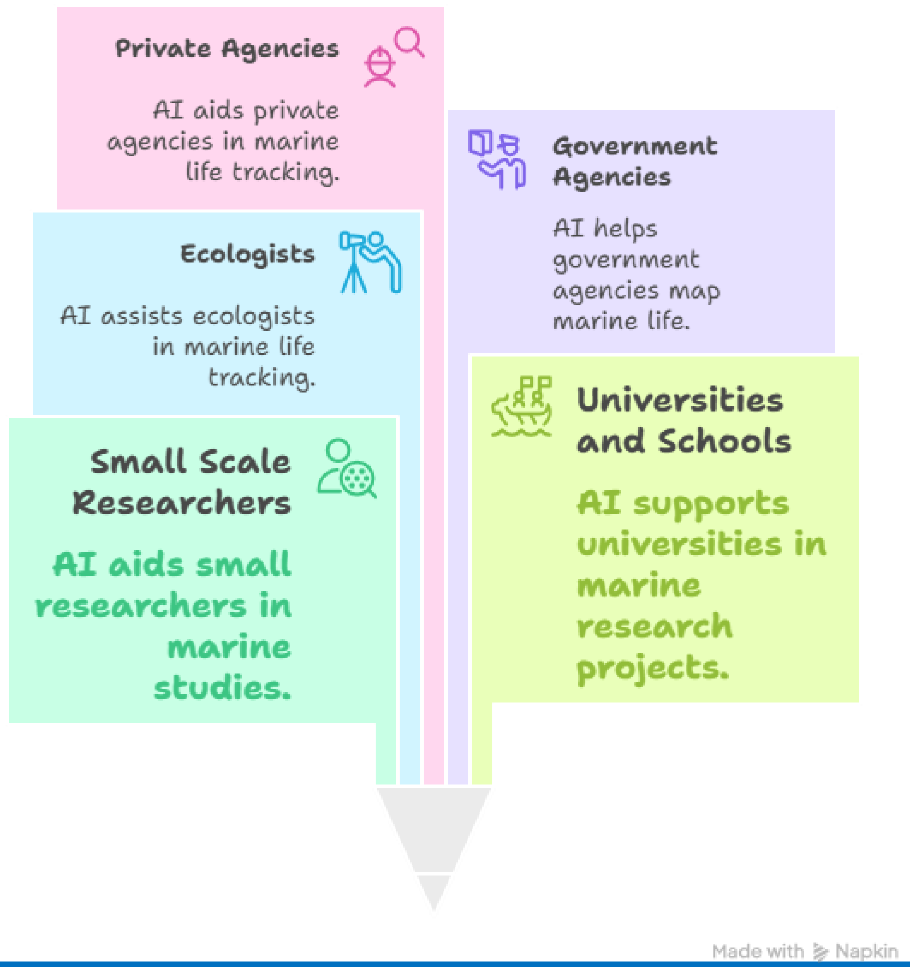
Our target audience for Oceanus AI are small scale researchers, universities and schools doing marine research, Ecologists and government and private agencies dealing in mapping and tracking marine life

It will help them save time and efforts by AI automating their classification work and novel species detection and with the additional features listed earlier also would let them get recognized and share their work to public on a platform dedicated to them



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RESEARCH AND REFERENCES

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Dataset collection:

- https://v3.boldsystems.org/index.php/Public_BINSearch?searchtype=records
- https://ftp.ncbi.nlm.nih.gov/blast/db/

Research websites:

- https://www.mdpi.com/1424-2818/12/4/123?utm_source=chatgpt.com
- https://link.springer.com/article/10.1007/s00227-023-04205-4?utm_source=chatgpt.com
- https://www.mdpi.com/2077-1312/12/6/971?utm_source=chatgpt.com
- https://ui.adsabs.harvard.edu/abs/2021BiCon..30.3087M/abstract

Our Website:

- https://sih-project-uimw.vercel.app/
- Backend deployment for API endpoint: https://sih-project-viay.vercel.app/